CSE-4301
Object Oriented Programming
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Week-2

Introduction to Object-Oriented Approach

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Outline

- Introduction to Object-Oriented Programming (OOP).
- Limitation of Procedural Paradigm.
- Thought Process of Object-Oriented Paradigm.
- Advantage of Object-Oriented Paradigm.
- Three Key Features of OOP

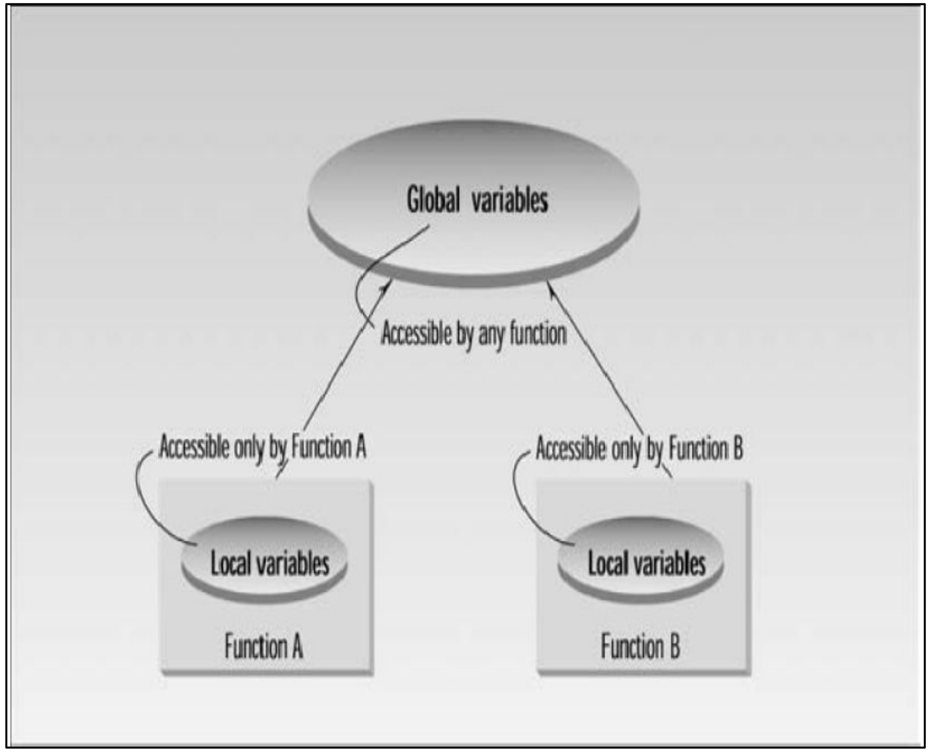
Introduction to OOP

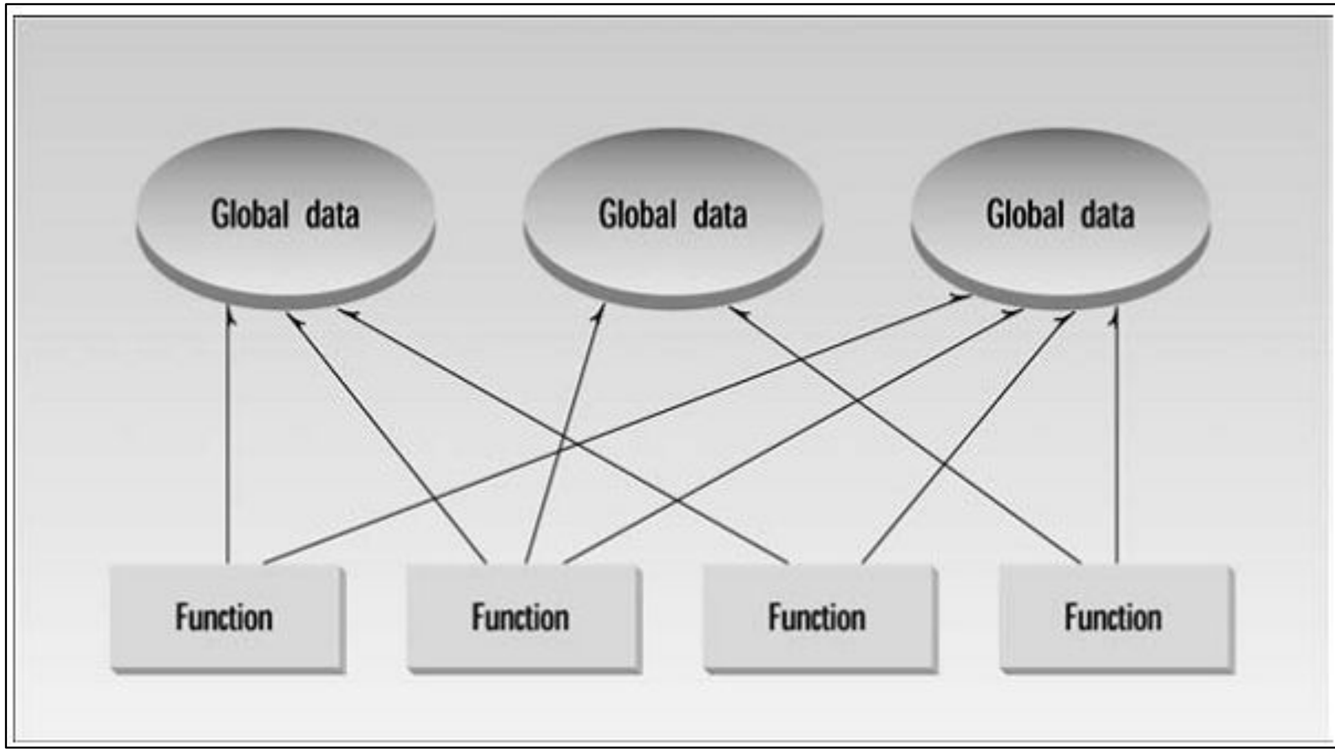
- Why do we need OOP?
- What does it do that traditional languages such as C, Pascal, and BASIC don't?
- What are the principles behind OOP?

Why Do We Need Object-Oriented Programming?

- Organizing Principles of Procedural Languages
  - Small program
    - list of instructions.
    - no other organizing principle (**paradigm**) is required.
  - Large Program
    - Division into Functions (subroutine, subprogram, or procedure)
    - function has a clearly defined purpose and a clearly defined interface
    - Cornerstone of Structured Programming.
- Problems with Structured Programming
  - Unrestricted Access
  - Real-World Modeling

Limitations of Structured Programming (Unrestricted Access)



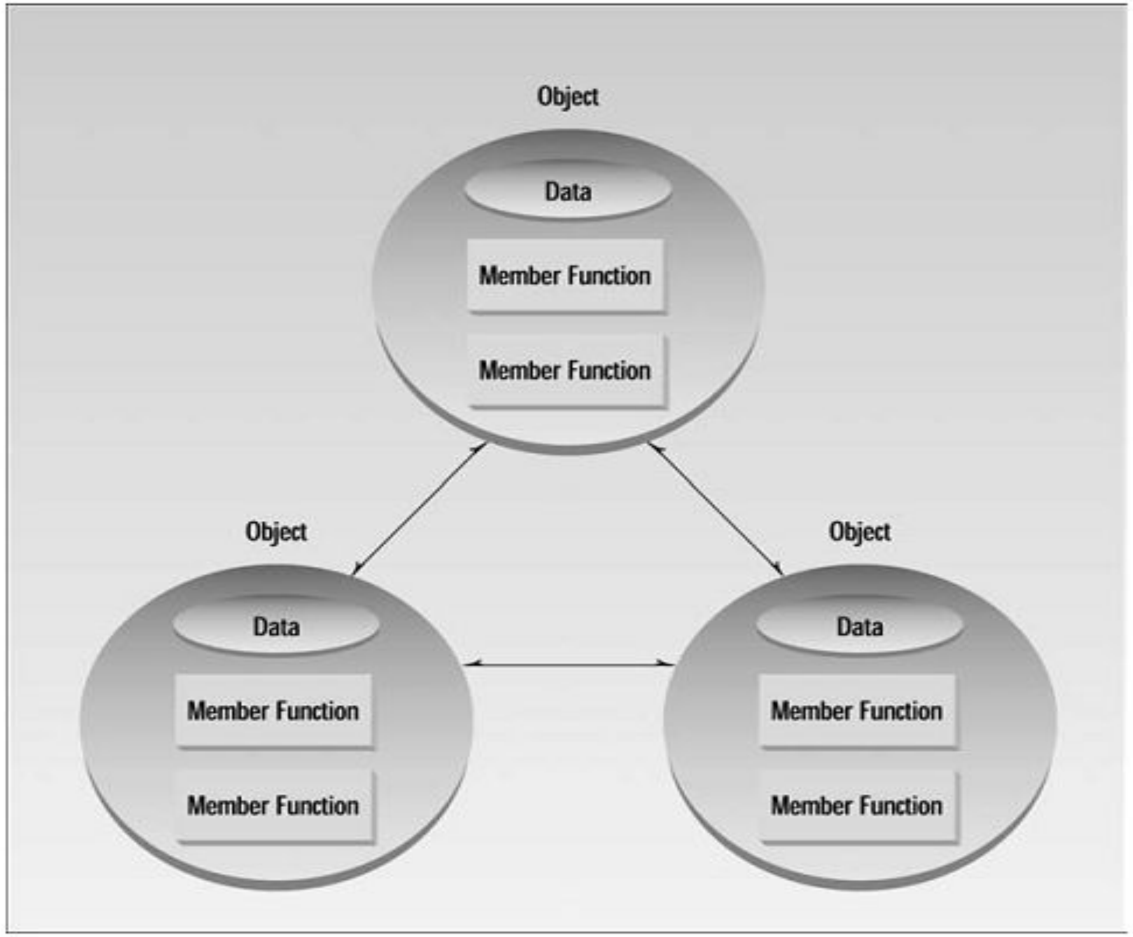


- Larger number of potential connections between functions and data
  - Program’s structure difficult to conceptualize
  - Difficult to modify

Limitations of Structured Programming (Real-World Modeling)

- Separation of Variable & Function in Structured Programming
  - Related variables and functions may not be in the same place (no organizing rule present)
  - Difficult for a new programmers to find all the functionalities and all the data present in the program for a real world's object.
  - Difficult to conceptualize.
- Real world object has **attribute** and **behavior**.
  - Attributes are the information/data of that object.
  - Behaviors are the functionalities of the object.

The Object-Oriented Paradigm

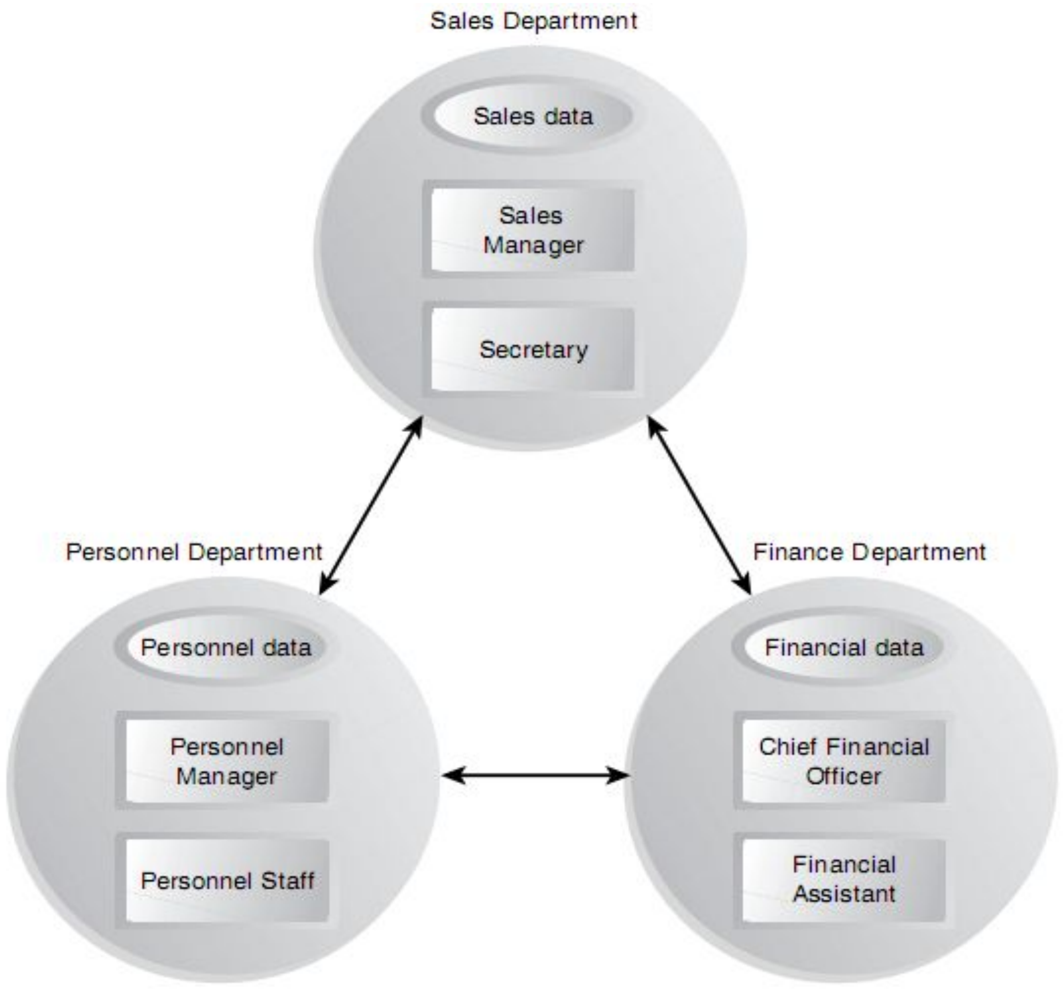


- An object in a program is a single unit that combines ***data*** and the ***functions*** that operate on that data.

Attribute -> Data -> **member variable**

Behavior ->Function -> **member functions**

An Analogy



- Each department has its specific role, data, assigned personnel, and clear division of responsibilities.
- Each department is responsible for handling its own data. Controlled access ensures data accuracy and prevents corruption.
- Objects provide a structured approach to organize programs which offers similar advantages.

INTRODUCTION to OOP

- OOP is a powerful way to approach the task of programming.
- OOP encourages developers to decompose a problem into its constituent parts.
- Each component becomes a self-contained object that contains its own instructions and data that relate to that object.

Characteristic of OOP language

- **Thinking in terms of objects, rather than functions**, has a surprisingly helpful effect on how easily programs can be designed.
- Match between **objects in the programming sense** and **objects in the real world**
- **Identification of Objects !!**

  [Page-17 of Object-Oriented Programming in C++ [Fourth Edition] -Robert Lafore**]**

Objects in Real world

- **Physical objects**
  - Automobiles in a traffic-flow simulation
  - Electrical components in a circuit-design program
  - Countries in an economics model
  - Aircraft in an air traffic control system
- **Elements of the computer-user environment**
  - Windows
  - Menus
  - Graphics objects (lines, rectangles, circles)
  - The mouse, keyboard, disk drives, printer

Objects in Real world

- **Data-storage constructs**
  - Customized arrays
  - Stacks
  - Linked lists
  - Binary trees
- **Human entities**
  - Employees
  - Students
  - Customers
  - Salespeople

Objects in Real world

- **Collections of data**
  - An inventory
  - A personnel file
  - A dictionary
  - A table of the latitudes and longitudes of world cities
- **User-defined data types**
  - Time
  - Angles
  - Complex numbers
  - Points on the plane

Objects in Real world

- **Components in computer games**
  - Cars in an auto race
  - Positions in a board game (chess, checkers)
  - Animals in an ecological simulation
  - Opponents and friends in adventure games

Advantages of OOP

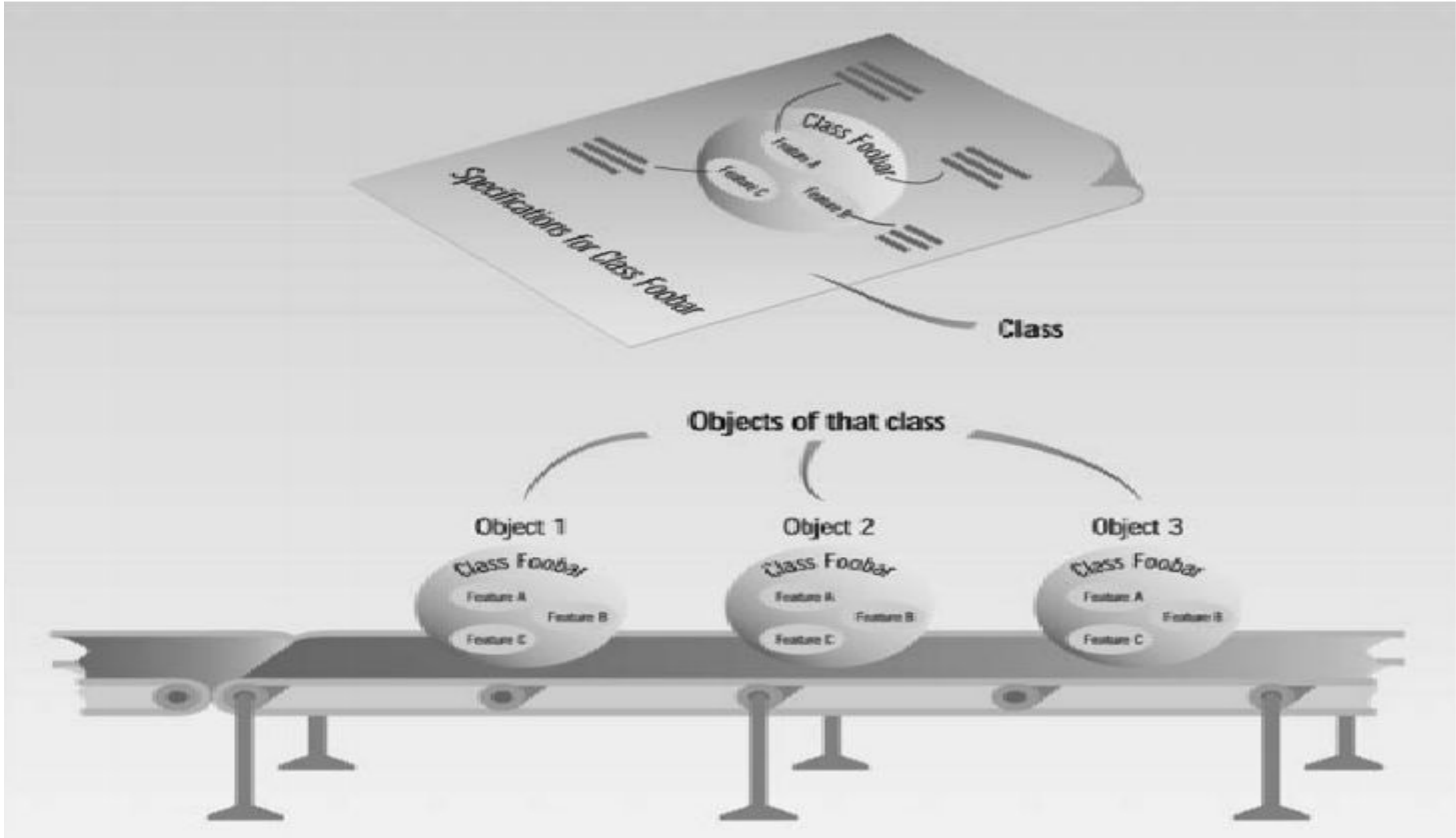
- Easy to Modify
- Easy to Conceptualize
- Easy to map Real world problem
- Programmer can manage larger programs
- Better Reusability Scope
- Complexity is reduced

What is a Class & Object ?

- A class is thus a
  - **description of several similar objects**
  - **Blueprint of an object**

- An object is often called
  - an **instance of a class**.
  - Single programming unit combining attributes and behaviors.
- Ada Lovelace, Alan Turing, Donald Knuth, and Dennis Ritchie are instance of the **ComputerScientist class**.

What is a Class & Object ?



Introduction To C++

- C++is the C programmer's answer to Object-Oriented Programming (OOP).
- C++ is an enhanced version of the C language.
- C++ adds support for OOP without sacrificing any of C's power, elegance, or flexibility.
- Both object-oriented and **non-object-oriented** programs can be developed using C++.
- C++ was invented in **1979 by Bjarne Stroustrup at Bell Laboratories** in Murray Hill, New Jersey, USA.

OOP Feature : C++

- All OOP languages, including C++, share three common defining traits:
  1. Encapsulation
     - Bundling of data (variables) and methods (functions)
  2. Polymorphism
     - Allows one interface, multiple methods
     - overriding and overloading
  3. Inheritance
     - Provides hierarchical classification
     - Permits reuse of common code and data

Reading Assignment

- Structure Detail from C
- Chapter-1 The Big Picture (Page- 10-27) of

  Object-Oriented Programming in C++ [Fourth Edition] -Robert Lafore
- Chapter-1 (Page- 1-9) of

  Teach Yourself C++ [Third Edition] –Herbert Schildt